FujiNet for the Fairchild Channel F, Rev0: an RP2040 Videocart on the F8 bus + ESP32-S3.

Signals flow left to right, from the console's cartridge slot to the network, and
every connection between the sheets is a wire on this page:

cart-rp	the 22-contact edge -> two SN74LVC4245A (5 V console / 3.3 V RP) -> the RP2040 (one symbol): D0-D7, ROMC0-4, WRITE, PHI, DIR; /INTREQ's open-drain FET; the activity LED; scope pads on DIR, WRITE, PHI, GP17; and the RP2040's own circuit: supplies and decoupling, 1.1 V core regulator, 2 MB QSPI flash, 12 MHz crystal, SWD, RUN / BOOTSEL / RESET, USB out to the S3
fujinet	the ESP32-S3 (FujiNet, the RP's USB host), microSD, WS2812 status LED
usb	USB-C, the CP2102N bridge and esptool's auto-program pair
power	+5V = console 5V OR USB VBUS (SS34s), the 3.3 V buck (+3V3), the RP's LDO (+3V3_RP)

Rails (GND, CONS_5V, VBUS, +5V, +3V3, +3V3_RP, DVDD) join by their power symbols.

Firmware contract (tools/check_nets.py checks this netlist against it):
fujinet-firmware pico/channelf channelf_cart.h (D0-D7 GPIO0-7, ROMC0-4 GPIO8-12,
WRITE GPIO13, PHI GPIO14, DIR GPIO15, /INTREQ GPIO16), boards/pico.h (LED GPIO25),
and the ESP32-S3 pin map fujiversal-channelf.h -- the RP2040 firmware runs unchanged.

Not verifiable in CAD (bring-up checklist, README): the WRITE edge and write-data
timing (PROVISIONAL in channelf_cart.c: scope pads), console 5 V headroom under
WiFi bursts, the Videocart edge's geometry (to be measured from a real cart).

power

File: power.kicad_sch

fujinet

File: fujinet.kicad_sch

cart-rp

USB_DP
USB_DM
RUN_CTL
BOOTSEL_CTL
S3_EN

File: cart-rp.kicad_sch

the console cartridge slot
(the edge) is on cart-rp

usb

S3_EN
S3_IO0
S3_TXD
S3_RXD

File: usb.kicad_sch

Firmware: fujinet-firmware pico/channelf (as-is) + fujiversal-channelf
CERN-OHL-W-2.0 (S3/USB/power from FujiNet-Astrocade-Rev0 and its sources)
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FujiNet

Sheet: /
File: FujiNet-ChannelF-Rev0.kicad_sch

Title: FujiNet Channel F Rev0

Size: A3	Date: 2026-10-09	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 1/5

The Fairchild Channel F Videocart edge, 22 contacts (channelf.se wiki Pinouts): the F8 bus has NO address bus -- D0–D7, five ROMC lines, WRITE (one machine cycle per pulse) and PHI -- and no reset line (a console reset arrives as ROMC 00). +12V (22) and NC (21) are left open.
U8 / U9 SN74LVC4245A: A port on CONS_5V (TTL inputs, VIH 2.0 V; drives 5 V levels -- the F3850 data bus wants VIH 2.9 V, Table 7). B port on +3V3_RP (the RP2040 is not 5 V tolerant). DIR high = A -> B = console -> cart.
U8, data: DIR = GPIO15 (channelf_cart.h: 1 = console -> cart, 0 = cart drives D0–D7). R18 holds it high through reset and boot against the pad pull-down. U9, control: DIR tied high. CONS_5V comes straight off the edge, ahead of the OR diode: with the console off both translators go high-Z (VCC isolation), so a USB-powered cart cannot back-power the console. /INTREQ (GPIO16, never driven) is open drain through Q1.

Supplies (Hardware design with RP2040, 2.1): IOVDD x6, USB_VDD, ADC_AVDD and VREG_VIN on +3V3_RP, the AP2112K LDO; the on-chip regulator makes DVDD (1.1 V) at VREG_VOUT for both DVDD pins. 100 nF at every supply pin, 1 uF at VREG_VIN and at VREG_VOUT (2.1.3). Crystal: ABM8-272-T3, 15 pF each side, 1k on XOUT (2.3). Flash: W25Q16JV, 2 MB (fujichannelf.cmake), /CS pulled up (2.2). USB: 27R at the chip (2.4); the S3 is the host. RESET resets the RP2040 and the ESP32-S3 together (D2 steers RST_BTN onto RUN and S3_EN) -- the way back to CONFIG after a booted Videocart. Hold BOOTSEL while pressing RESET for the RP's USB boot ROM; the S3 does both itself: IO4 (RUN_CTL) low resets the RP, IO5 (BOOTSEL_CTL) low through that reset selects BOOTSEL. First flash: SWD pads (the RP has no PC-facing USB).

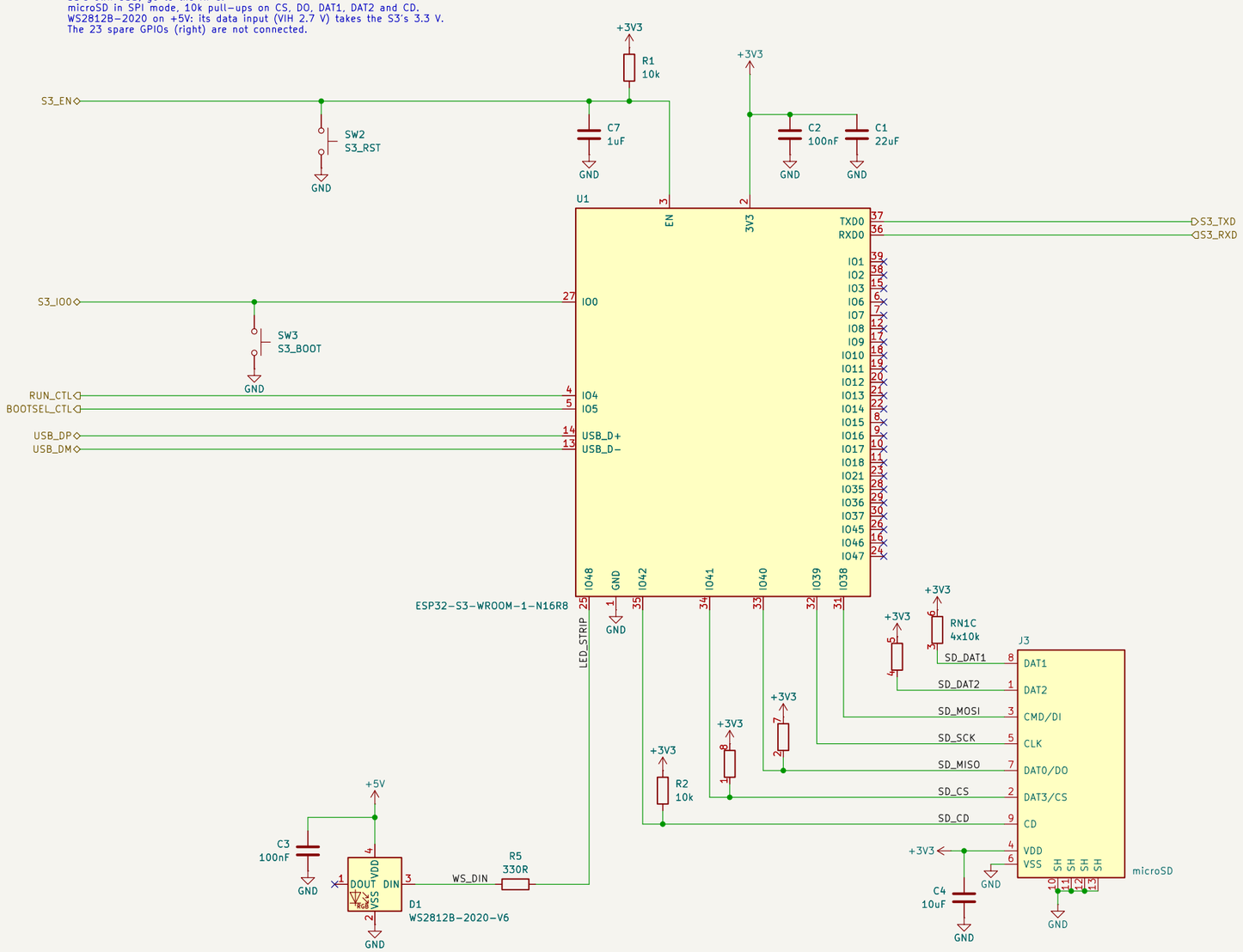
Firmware: fujinet-firmware pico/channelf (as-is) + fujiversal-channelf
CERN-OHL-W-2.0 (S3/USB/power from FujiNet-Astrocade-Rev0 and its sources)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file
FujiNet

Sheet: /cart-rp/
File: cart-rp.kicad_sch

Title: FujiNet Channel F Rev0 — Cart bus -> RP2040

Size: A2	Date: 2026-10-09	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 2/5

ESP32-S3-WROOM-1-N16R8: the FujiNet (fujiversal-channelf.h pin map).
EN: 10k / 1 uF power-on delay. The S3_RST button, the cart RESET
(cart-rp, through the BAT54C) and the USB bridge's auto-program pair
(usb sheet) all pull it low; S3_BOOT and the same pair pull IO0.
IO4 / IO5 drive the RP2040's RUN and BOOTSEL; IO19 / IO20, the
S3's own USB, go to the RP's.
microSD in SPI mode, 10k pull-ups on CS, DO, DAT1, DAT2 and CD.
WS2812B-2020 on +5V: its data input (VIH 2.7 V) takes the S3's 3.3 V.
The 23 spare GPIOs (right) are not connected.



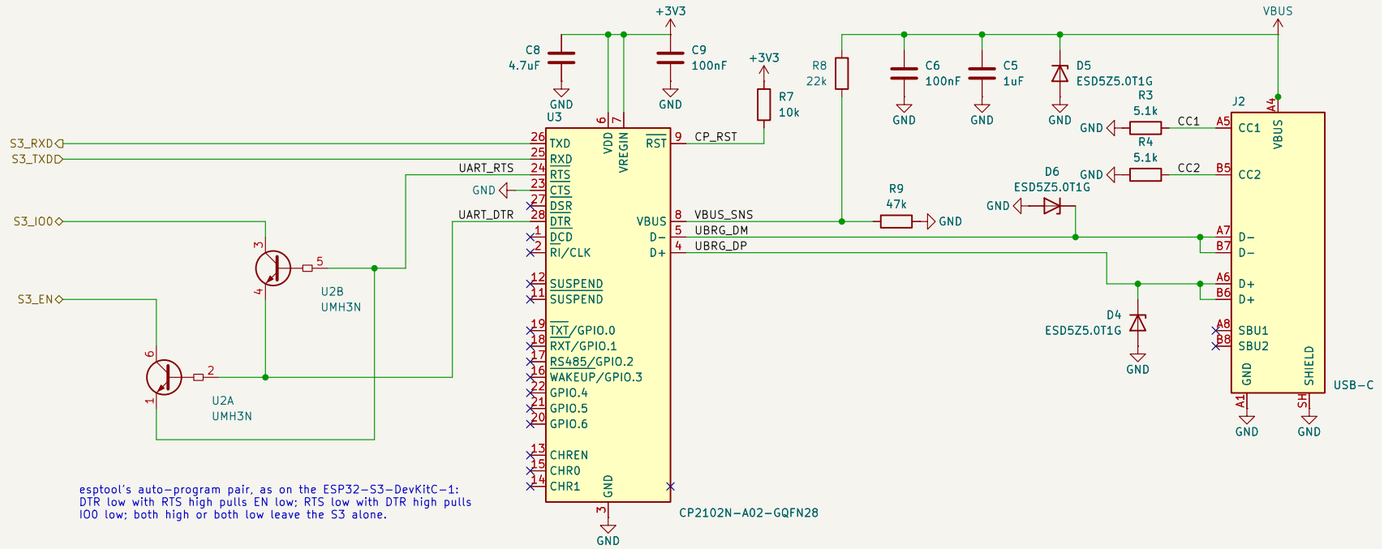
Firmware: fujinet-firmware pico/channelf (as-is) + fujiversal-channelf
CERN-OHL-W-2.0 (S3/USB/power from FujiNet-Astrocade-Rev0 and its sources)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

FujiNet

Sheet: /fujinet/
File: fujinet.kicad_sch

Title: FujiNet Channel F Rev0 - FujiNet (ESP32-S3)

Size: A3	Date: 2026-10-09	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 3/5



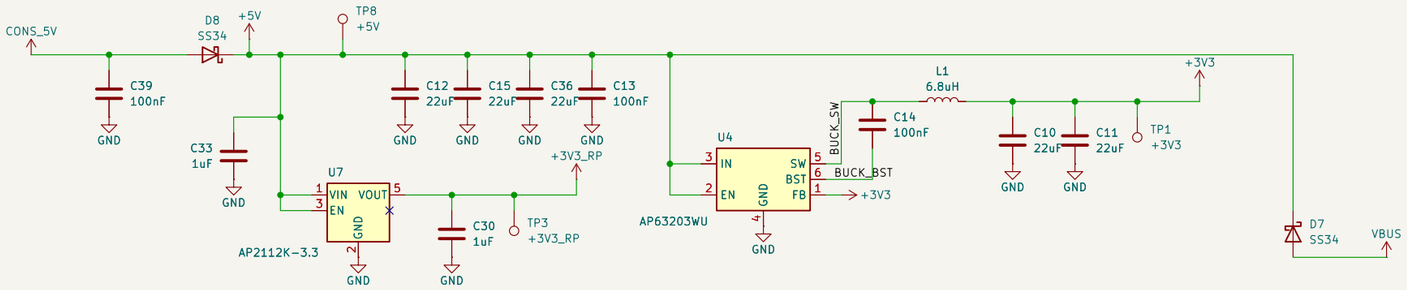
Firmware: fujinet-firmware pico/channelf (as-is) + fujiversal-channelf
CERN-OHL-W-2.0 (S3/USB/power from FujiNet-Astrocade-Rev0 and its sources)
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FujiNet

Sheet: /usb/
File: usb.kicad_sch

Title: FujiNet Channel F Rev0 — USB-C + CP2102N

Size: A3	Date: 2026-10-09	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 4/5

Power tree: +5V is the OR of two sources, so neither can back-feed the other.
  CONS_5V edge pins 19 / 20, the console's +5V, through D8 SS34.
  VBUS   USB-C, through D7 SS34.
SS34 VF ~0.35 V at the cart's ~0.4 A: +5V is 4.4-4.9 V. The only 5 V logic is the two SN74LVC4245A A ports (cart-rp sheet), and they take CONS_5V itself, ahead of the diode: off with the console, so the translators isolate the bus whenever the console is off.
U7 AP2112K-3.3 LDO -> +3V3_RP: the RP2040 (IOVDD, USB_VDD, ADC_AVDD, VREG_VIN), its flash and both translators' B ports; in dropout it follows +5V up as the console rail rises.
U4 AP63203 3.3 V / 2 A buck -> +3V3: ESP32-S3 (WiFi bursts ~400 mA), CP2102N, microSD; it waits for UVLO, then soft-starts over 4 ms. TP8, TP1 and TP3 are bring-up test points. Console +5V headroom under WiFi bursts is unmeasured: power the cart from USB-C if the console rail sags (bring-up checklist, README).



PWR_FLAG: these rails reach their loads only through passive parts (the edge pins, USB-C, the SS34s, the buck's inductor).



Firmware: fujinet-firmware pico/channelf (as-is) + fujiversal-channelf
CERN-OHL-W-2.0 (S3/USB/power from FujiNet-Astrocade-Rev0 and its sources)
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FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet Channel F Rev0 - Power

Size: A3	Date: 2026-10-09	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 5/5